

Syllabus

Spring semester 2025

Course Title	Chemistry and Art	Course Code		KC6101		Credit	3	Class room	hyung3303, chang610
		Type		elective		Hours	3		
Teacher	Name	Class no.	Office	Office hours	Contact		E-mail		
	Choi, Eun-Young	1	Hyung 3303	M8, Tu8, F7	606-2389		faujasite1@kaist.ac.kr		

1. Course Description

This course encourages students to explore the intersection of chemistry and art through the study of conservation science, crystal visualization, and coloring materials. Students will engage with both chemistry and aesthetics by creating and analyzing crystal structures and frameworks. They will also explore the significance of color and experiment with producing dyes and pigments using metal oxide chemicals. Additionally, students will investigate the relationship between structure and color to refine the quality and texture of pigments. Ultimately, they will have a unique opportunity to create artwork using their own handcrafted pigments."

2. Core Target Competencies

Subject Core Competencies						
. Analytical and Creative Thinking ¹⁾	Complex Communication ²⁾	Leadership and Teamwork ³⁾	. Adaptability/ Progression / Spirit of Adventure ⁴⁾			
1) Ability to find quality information in the flood of information to articulate meaningful questions, to analyze and create ideas or knowledge 2) Ability to communicate clearly with diverse audiences or readers, to communicate clearly with diverse audiences or readers, to listen attentively 3) Ability to be the first to come up with and present new ideas, tasks, manage groups, and delegate responsibilities 4) Ability to explore and experiment						

3. Textbooks and References

Literature Papers

4. Class Method

A. Lecture

- A Class that stimulates active student participation

B. Debate/Discussion

C. Activities

- Practical exercises utilizing programming and database applications

- Creative chemistry and art activity

5. Evaluation

A. Evaluation Criteria

Final-term	Project Presentation	Report & Exhibition	Attendance	Attitude	Total*
30	30	20	10	10	100%

B. Evaluation Method

(1) The grade will be given according to total points.

(2) Peer review will be included for the grade.

(3) Attendance and Attitude

- No absences and No late arrivals Full score (10 points)
- Being late once: -0.2 points.
- One, two, and three absence: -1, -2, and -5
- Students with more than four unexcused, full-class absences will receive an "F" grade for the course.
- Using electric device without permission (e.g., chatting on line, playing games): -10 points.

6. Criteria for Awarding Credits

☛ Absolute Evaluation

☛ Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.

7. Weekly Plans

Week(period)	Topic and Contents	(Notes)
Week 1 2/17-2/21	Orientation Introduction to chemistry and Art -Installation CSD&Mercury; Artwork Conservation Chemistry	Lecture, activity
Core Competencies	♦ Complex communication	
Week 2 2/24-2/28	Learn and practice how to use visualizer programs Visualization of Crystalline materials	Lecture, activity
Core Competencies	♦ Analytical and Creative thinking, Leadership and Teamwork	
Week 3 3/3~3/7	1) Art and the integration of chemical frameworks: A literature review 2) Tessellation 3) Visualization, simplification, and Aesthetic representation of materials	Lecture, activity
Core Competencies	♦ Analytical and Creative thinking, Complex communication	
Week 4 3/10~3/14	Presentation of literature studies (Artwork Conservation Chemistry including dye and Pigments)	
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Leadership and Teamwork	
Week 5 3/17~3/21	Presentation of literature studies (Artwork Conservation Chemistry including dye and	activity

	pigments)	
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Spirit of Adventure	
Week 6 3/24~3/28	Journal cover design based on crystalline materials you created using a visualization program.	Lecture, Activity
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Leadership and Teamwork	
Week 7 3/31~4/4	The Journal cover exhibition	Lecture, Activity
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Leadership and Teamwork	
Week 8 4/7~4/11	Midterm Exams	
Week 9 4/14~4/18	Create Red, White, Blue Oil Paint using Metal Oxide1	Activity
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Spirit of Adventure	
Week 10 4/21~4/25	Create Red, White, Blue Oil Paint using Metal Oxide2	Experiment
Core Competencies	♦ Analytical and Creative thinking, Spirit of Adventure, Complex communication	
Week 11 4/28~5/2	Create dye and pigments using chemicals	Experiment
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Leadership and Teamwork	
Week 12 5/5~5/9	Analysis of Pigments	
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Leadership and Teamwork	
Week 13 5/12~5/16	Comparison of your pigments and commercial ones	
Core Competencies	♦ Analytical and Creative thinking, Spirit of Adventure , Leadership and Teamwork	
Week 14 5/19~5/23	Create artwork using pigments.	
Core Competencies	♦ Analytical and Creative thinking, Spirit of Adventure , Leadership and Teamwork	
Week 15 5/26~5/30	Contemporary Art Exhibition	
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Leadership and Teamwork	
Week 16 6/9~6/13	Final Exams	

※ The above plans may change according to the academic schedule.

[Reference] Syllabus Competence Reference

1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/ Quantitative Literacy	5. Global Perspective	6. Adaptability/ Progression / Spirit of Adventure	7. Integrity and Ethical Decision Making	8 Habits/ Mindset
Ability to identify, manage and deal with complex problems	Ability to understand and express themselves in two or more languages	Ability to be the first to come up with and present new ideas	Ability to understand, use and apply digital technology	Ability to develop an open mind about the traditions or values of others	Ability to develop flexibility, agility and adaptability	Ability to sustain an empathetic and understanding perspective	Sincerity
Ability to see through prejudice and identify reliable information and uncertain information	Ability to communicate clearly with diverse audiences or readers	Ability to influence people to follow themselves, rather than lead them through directives, orders, or authority	Ability to create digital contents and media	Ability to understand the history, politics, religion, and culture of another region	Ability to challenge unfamiliar situations	Ability to cultivate sincerity, honesty, fairness, and respect	Creativity
Ability to find quality information in the flood of information	Ability to listen attentively	Ability to build trust, resolve conflicts, and support others	Ability to utilize multimedia resources to effectively communicate one's ideas in a variety of formats	Ability to use one or more of the international languages used worldwide	Ability to explore and experiment	Ability to display moral courage in the face of injustice	A love of learning and curiosity
Ability to articulate meaningful questions	Ability to speak effectively and coherently	Ability to facilitate group discussions or discussions, build consensus, and negotiate outcomes	Ability to master and use higher mathematics	Ability to use technology to connect people and things around the world	Ability to word effectively in situations of ambiguous or changing priorities	Ability to act responsibly for the benefit and happiness of the greater community	Resilience/elasticity
Ability to analyze and create ideas or knowledge	Ability to write concisely and clearly, considering the diversity of readers	Ability to teach, guide, and counsel Ability to seek help from others	Ability to understand traditional or recently emerging fields such as mathematics, science, technology, environmental science, robotics, geometry, cellular automata, nanotechnology, and biotechnology	Social and intellectual skills to effectively navigate information across multiple cultures	Ability to see failure as a lesson and to accept that innovation is made up of frequent failures and small successes	Ability to have an intrinsic understanding of ethical issues or dilemmas about new media or new technologies	Perseverance/patience
Ability to	Ability to	Ability to		Ability to use 21 st	Ability to develop	Ability to make	Self-efficacy

1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/ Quantitative Literacy	5. Global Perspective	6. Adaptability/ Progression / Spirit of Adventure	7. Integrity and Ethical Decision Making	8 Habits/ Mindset
devise and test solutions to problems by trial and error	explain information and persuasively engage others in an engaging way	collaborate on tasks, manage groups, and delegate responsibilities		century core competencies such as communication skills, collaboration skills, critical thinking skills, creativity to properly understand and deal with global issues	a sense of independence to seek new roles, ideas, or strategies	rational and ethical decisions on complex issues	
Ability to think of different ways to cope		Ability to listen and practice goals		Ability to learn from and collaborate with people from different cultures, religions and lifestyle through mutual respect and open dialogue	Ability to develop the knowledge and competencies necessary to start a business and run a business		Stress management skills
Ability to develop knowledge and perspectives that are not limited to one field but span multiple fields		Ability to share credit for success		Ability to generate diverse ideas and exploit social and cultural differences to get things done			Time management skills
Ability to make sustained inferences							
Ability to put things together or adapt them to new uses and situations							
Ability to solve new problems that cannot be solved through established rules							
Ability to use knowledge or creativity to solve complex real-world problems							

1. Complex Problem-Solving Skills	Ability to solve new and unclear problems that may arise in complex real-world situations
2. Critical Thinking Skills	Ability to compare and analyze the strengths and weaknesses of various solutions or approaches to a given problem using logic and reasoning
3. Creativity	Ability to come up with novel ideas when given a certain topic or situation or to suggest a creative method in the process of solving a problem
4. Human Resource Management Skills	Ability to select the most suitable person for a specific task and manage them to achieve maximum work performance through motivation, and systematic efficient management
5. Ability to Collaborate	Ability to adjust one's own behavior in response to the behavior of others
6. Emotional Ability	Ability to perceive other people's reactions and understand why they react the way they do
7. Judgement and Decision-making Skills	Ability to make the most rational judgement and decision by analyzing the costs and benefits of a specific situation or problem
8. Orientated Towards Service	Ability to actively seek ways to help others
9. Negotiation Skills	Ability to bring together parties who differ in ideas or interests and try to reduce those differences
10. Cognitive Flexibility	Ability to apply different kinds of rules or principles to rearrange and reorganize things in different ways